

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 414 — Past Questions & Materials
by Renance

Course: MEE 414
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — MEE 414 Metallurgy II

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM

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i 3 DEPARTMENT OF MECHANICAL ENGINEERING

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ee,

io) First Semester Examination 2024/2025 Academic Session

COURSE CODE AND TITLE: MEE 414- METALLURGYN IT

TIME ALLOWED: TWO (2) HOURS (for 2 Units)

INSTRUCTION: ANSWER THREE QUESTIONS: QUESTION 1 AND ANY OTHER

TWO

QUESTION 1- Compulsory

As regards the assigned group project, do the following:

- Briefly describe the production of the assigned non-ferrous metal in less than a page. (4 marks]
- State and explain its properties. [3 marks]
- List its uses/applications based on its properties. (3 marks]
- What was your role in accomplishing your group work? [7 marks]
- Please suggest any other use/application of the metal/alloy yet to be explored.

[2 marks]

f. Which of the properties do you con

[2 marks] .

g. Where/how is the material applicable i

marks]

h. Where does the material exist natura

sider to have dominance of effect over others? _-

n the oil and gas sector if possible? [2

lly in Nigeria/world? [2 marks]

QUESTION 2

- With the aid of a table, compare non-metals and metals using physical/mechanical properties but not limited to these: appearance, conductivity, malleability, ductility, state at room temperature, density. [5 marks]
- State and explain any five unique properties that make non-metals for high temperature applications if any. [10 marks]

c. Differentiate between the basic non-metallic engineering materials with the aid of their atomic structures? [10 marks]

Is suitable

QUESTION 3

a. Discuss and differentiate between the deterioration of each of these materials ~ metals, ceramics and polymers? [5 marks]

b. What are the best methods to prevent corrosion? [10 marks]

c. The oil and gas industry encounters several types of corrosion due to extreme environmental conditions, chemical exposure, and high-pressure operations. What type of corrosion is expected in each of the stated scenario and state the solution you will proffer for each case? [10 marks]

QUESTION 4

a. Metal fabrication techniques are normally preceded by refining, alloying and often heat-treating processes that produce alloys with the desired characteristics. With explanations suggest which is preferred in engineering applications, pure metals or alloys? [5 marks]

b. Differentiate between these two fabrication processes: metal forming and casting, by definition, indication of their common types and processes using diagram to simplify your answer. [20 marks]

c. Which of these phenomenon - plastic deformation and solidification principle is applicable to:

(i) metal forming

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(ii) casting process,

Please explain with diagrams where necessary. [10 marks]